

ForMoSA: Forward Modeling tool for Spectral Analysis

ForMoSA Collaboration^{*}, Simon Petrus^{1,2,3,4*}, Paulina Palma-Bifani^{5,6*},
Matthieu Ravet^{5,4,7*}, Allan Denis^{8*}, Bhavesh Rajpoot^{7,9*}, Arthur
Vigan^{8*}, Mickaël Bonnefoy^{4*}, Gaël Chauvin^{7*}, Alice Radcliffe^{6*},
Kevin Hoy^{2,3,10*}, and Pablo Requeijo^{6*}

¹ NASA-Goddard Space Flight Center, Greenbelt, MD 20771, USA ² Instituto de Estudios Astrofísicos,
³ Facultad de Ingeniería y Ciencias, Uni. Diego Portales, Av. Ejército 441, Santiago, Chile ⁴ Millennium
⁵ Nucleus on Young Exoplanets and their Moons ⁶ Univ. Grenoble Alpes, CNRS, IPAG, F-38000 Grenoble,
⁷ France ⁸ Laboratoire J. L. Lagrange, Université Côte d'Azur, Observatoire de la Côte d'Azur, CNRS,
⁹ 06304 Nice, France ¹⁰ LIRA, Observatoire de Paris, Université PSL, Sorbonne Université, Université de
¹¹ Paris, 5 place Jules Janssen, 92195 Meudon, France ¹² Max-Planck-Institut für Astronomie, Königstuhl
¹³ 17, 69117 Heidelberg, Germany ¹⁴ Aix Marseille Univ, CNRS, CNES, LAM, Marseille, France ¹⁵ Department of Physics and Astronomy, Heidelberg University, Im Neuenheimer Feld 226, D-69120
Heidelberg, Germany ¹⁶ European Southern Observatory, Alonso de Cordova 3107, Vitacura, Santiago,
Chile * These authors contributed equally.

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Editor: [Open Journals](#) 

Reviewers:

- [@openjournals](#)

Submitted: 01 January 1970

Published: unpublished

License

Authors of papers retain copyright
and release the work under a
Creative Commons Attribution 4.0
International License ([CC BY 4.0](#)).

Summary

ForMoSA (FORward MODELing tool for Spectral Analysis) is an open-source Python package designed to extract physical parameters from spectroscopic and photometric observations based on a Bayesian framework. It has been mainly designed for young planetary-mass objects and directly imaged exoplanets, and it can utilize different self-consistent atmospheric models to perform robust parameter space exploration.

Over the past years, **ForMoSA** has evolved from a specialized research script into a mature, community-driven tool. The developments within **ForMoSA** are supported by an international collaboration among several laboratories in France (IPAG, LIRA, LAM, and Lagrange), Germany (MPIA), USA (NASA Goddard), and Chile (FCLA, Universidad Diego Portales, and Universidad de Chile). This evolution has led to the need for this dedicated publication where we formally present the tool to the community.

The paper is organized as follows: the Statement of need section provides a statement of need; the Software design section describes the specific functionalities of the code; the Installation and example section points towards the online documentation; and the Performance section evaluates the performance of the code. Finally, the Conclusions section concludes with prospects on future directions for **ForMoSA**.

To complement the information in the main body, several appendices are provided: one introduces other community standard tools; another discusses the expected accuracy of the framework; and a third summarizes the peer-reviewed work utilizing **ForMoSA**.

Statement of need

Recent advances in ground- and space-based observatories now enable routine, high-quality observations of exoplanet atmospheres across a wide range of wavelengths and resolutions. The code **ForMoSA** was developed to address the growing need for an efficient, standardized framework to analyze these high-contrast, high-resolution observations coming from various

instruments. This includes the need to analyze multiple datasets taken at different times and with different setups within a single generalized framework.

In practice, **ForMoSA** was designed to bridge the gap between these observations and atmospheric models by providing a Bayesian framework to robustly compare the two. While several atmospheric inference codes exist, most rely on a data-driven retrieval approach where spectra are generated on the fly using highly parameterized models (Madhusudhan, 2018). In contrast, **ForMoSA** adopts a forward-modeling approach based on pre-computed, self-consistent atmospheric model grids.

This approach enables a direct model-driven comparison between observations and physically motivated theoretical predictions. Examples of other similar codes are given in Appendix A. In addition, by providing both a terminal-based interface and interactive Jupyter Notebook demos, **ForMoSA** stands out by the following key advantages.

- **Computational efficiency:** It leverages `xarray`¹ for optimized multi-dimensional array management, enabling fast interpolation and manipulation of large model grids.
- **Data modularity:** The framework allows for the simultaneous import and analysis of diverse datasets, including spectroscopic and photometric observations, even when they overlap in wavelength coverage.
- **Flexible likelihood mappings:** Users can choose from various likelihood metrics, including different noise and scaling prescriptions to account for specific observational uncertainties (Ruffio et al., 2019).
- **Extensive parameterization:** Beyond the standard model grid, a comprehensive suite of extra-grid parameters, such as radial velocity, rotational broadening, and scaling factors, can be incorporated into the fit.
- **High-Contrast module:** It incorporates the method of (Landman et al., 2024), allowing users to include reference stellar, atmospheric transmission, and systematics spectra to accurately fit contrast-limited data.
- **Active development:** It is regularly updated to incorporate the latest algorithmic advancements and new model versions.

Software design

The Python-based code of **ForMoSA** is organized into several modules, as depicted in Figure 1 and Tables 1 and 2. In practice, **ForMoSA** has a class-based architecture, with some main modules (`config`, `observation`, `filter`, `grid`, `parameter`, `nested_sampling`; see the right side of the workflow figure), and support modules (`core`, `utils`, and `transform`; see the left side of the workflow figure).

At the beginning, the user must input a grid of self-consistent, precomputed atmospheric models, along with a set of observations and a configuration file containing all of **ForMoSA**'s input parameters, including those for the nested sampling algorithms, the log-likelihood function, and the nested sampling parameters associated with their priors. Then, **ForMoSA**'s main modules are centralized through the analysis class, which creates and handles sub-grids of atmospheric models tailored to each observation, before launching the nested sampling algorithm using the support modules.

Later, at each iteration of the nested sampling, random parameter values are drawn from the prior distribution specified by the user for each parameter. The transformations related to these parameters are applied to the sub-grids, before comparing them to each observation by computing the log-likelihood function. For a given run, the data, sub-grids, parameters, and results are automatically saved to paths specified by the user, and can be easily recovered by

¹`xarray` documentation: <https://docs.xarray.dev/en/stable>

87 the modules of **ForMoSA** for subsequent analysis. The results can also be visualized through
88 the Plotting class.

ForMoSA workflow diagram. The shaded gray area on the right represents the core functionalities required to run the Nested Sampling. The left dark-gray area represents utility and support functions. The boxes contain the classes of modules of **ForMoSA**. The main methods of each class are depicted as subtext below each box. The larger dashed boxes represent the contents of each folder, while the smaller dashed boxes represent sub-modules. The larger boxes represent the main modules.

Figure 1: ForMoSA workflow diagram. The shaded gray area on the right represents the core functionalities required to run the Nested Sampling. The left dark-gray area represents utility and support functions. The boxes contain the classes of modules of **ForMoSA**. The main methods of each class are depicted as subtext below each box. The larger dashed boxes represent the contents of each folder, while the smaller dashed boxes represent sub-modules. The larger boxes represent the main modules.

89 **Table 1: Main modules of ForMoSA**

Module	Description
config	
ConfigGenerator (global_config.py)	Class handling the generation and saving of a default configuration file.
ConfigLoader (global_config.py)	Class handling the loading of parameters from a configuration file.
Paths (paths.py)	Class handling the paths used in the configuration file.
observation	
Observation (observation_base.py)	Base class representing an observation (spectroscopic or photometric).
SpectralObservation (observation_spectroscopy.py)	Class handling a spectroscopic observation.
PhotometryObservation (observation_photometry.py)	Class handling a photometric observation.
ObservationSet (observation_set.py)	Container for a set of observations.
ObservationLoader (observation_set.py)	Generate observation objects from various input types.
filter	
PhotometryFilter (filter.py)	Class defining a photometric filter.
grid	
ModelGrid (model_grid.py)	Class representing a native (non-adapted) grid.
SubGrid (subgrid_base.py)	Base class representing a subgrid (spectroscopic or photometric).
SubGridSpectroscopy (subgrid_spectroscopy.py)	Class handling a spectroscopic subgrid.
SubGridPhotometry (subgrid_photometry.py)	Class handling a photometric subgrid.
SubGridSet (subgrid_set.py)	Container for a set of subgrids.
GridLoader (grid_loader.py)	Generate grid from various input types.
parameter	
Parameter (parameter.py)	Class handling a single parameter.

Module	Description
ParameterSet (parameter_set.py) Prior (prior.py)	Container for a set of parameters. Base class representing a prior associated to a parameter.
nested_sampling NestedSampling (nested_sampling.py) NSResults (results.py) NSAnalysis (ns_analysis.py) Plottings (plotting.py)	Class handling the Nested Sampling. Class handling the results of the Nested Sampling. Class for post-analysis of the Nested Sampling results, including reconstruction of the model from the best fit and the confidence intervals. Class handling the visualization of the results of the Nested Sampling.

⁹⁰ **Table 2:** Support modules of **ForMoSA**

Module	Description
transform ApplyPhysicsEffects (apply_effects.py) ApplyObservationEffects (apply_effect.py) ObservedParameters (observed.py) ObservedModel (observed.py) PhotometricEffects (photometric_effects.py) SpectroscopicEffects (spectroscopic_effects.py)	Apply physics-based transformations to a model. This includes, for example, Doppler shift, rotational broadening, radius scaling. Apply observation-based transformations to a model. This includes resolution decreasing, analytic scaling, or high-contrast modeling. Class defining a set of parameters and their associated values drawn from the Nested Sampling at a given iteration. Class defining the model drawn from the Nested Sampling at a given iteration. Class defining the physics-based and observation-based transformations associated to photometry. Class defining the physics-based and observation-based transformations associated to spectroscopy.
core config.py enums.py errors.py loggings.py	General configuration setup and examples for plotting and saving. Enumerations of various keys used in ForMoSA . Exception class. Logging module.
utils logL_functions.py misc.py prior_functions.py spec.py	Definition of loglikelihood functions used in ForMoSA . Miscellaneous utilities. Definition of priors used in ForMoSA . Support functions for the transformation of spectrophotometric data.

Installation and example

Because **ForMoSA** is an end-to-end Bayesian tool that handles all stages of the analysis, such as adapting model grids to the data format, performing the Bayesian inversion, and generating the results, its installation requires several Python packages, as well as downloading the model grids needed for the fit. All information required for installation, access to available grids, and the demonstration material is available at the **ForMoSA** documentation².

Some of the most commonly used model grids are available in a **ForMoSA**-compatible format. If a specific model required for an analysis is not yet available, the corresponding grid can be generated and reformatted by the developer upon request. Finally, to facilitate onboarding and training of new users, demonstration notebooks and tutorials are provided.

Performance

The computational cost of **ForMoSA** is primarily driven by the number of forward model evaluations required for the nested sampling algorithm to converge. While absolute execution time (inversion time) depends on the specific hardware, this section provides a comparative analysis to guide users in defining the configuration file.

As astrophysicists, even though computing-time optimization is useful and important, we are in practice primarily driven by the retrieval accuracy. We complement this analysis by evaluating the accuracy for a specific test case in Appendix B.

Performance comparison between the nested sampling algorithms PyMultiNest (squares) and Nestle (crosses). Different colors show the number of free parameters used (from 1 to 5). From left to right: Inversion time as a function of spectral resolution (R_λ), signal-to-noise (S/N), and number of live points. The default setup is $R_\lambda = 368$, S/N = 22, and 215 live points. This is intended to inform the user of the order of magnitude in time they should expect for their fit to converge.

Figure 2: Performance comparison between the nested sampling algorithms PyMultiNest (squares) and Nestle (crosses). Different colors show the number of free parameters used (from 1 to 5). From left to right: Inversion time as a function of spectral resolution (R_λ), signal-to-noise (S/N), and number of live points. The default setup is $R_\lambda = 368$, S/N = 22, and 215 live points. This is intended to inform the user of the order of magnitude in time they should expect for their fit to converge.

The total inversion time is a multi-dimensional function depending on several factors, including the spectral resolution (R_λ), the signal-to-noise ratio (S/N), the number of live points, the dimensionality of the parameter space, and the machine used. Figure 2 illustrates how these various parameters scale the inversion time using synthetic observations.

In the left panel, we observe that the main driver of computational cost is indeed the spectral resolution. As the spectral resolution increases, the number of data points also increases, and the number of points to be compared in the likelihood function grows, directly increasing the computational cost of each forward model call.

In the middle panel, we observe the dependence on the S/N. Here, noise is applied using the following formula:

$$\vec{n} \sim \mathcal{N}(0, \sigma^2), \quad \text{with} \quad \sigma = \frac{\text{mean}(d)}{S/N},$$

where d is the synthetic flux. As the S/N increases, the error bars become smaller, which leads to a more sharply peaked likelihood function. This tighter posterior distribution requires the

²**ForMoSA** documentation: <https://ForMoSA.readthedocs.io/en/latest/index.html>

sampler to explore the parameter space with finer precision and typically increases the number of likelihood evaluations needed for convergence, thereby lengthening the total inversion time.

In the right panel, we observe that the inversion time scales approximately linearly with the number of live points. It is therefore important to consider that while more points improve the sampling of the posterior, they significantly increase the time required to reach the termination criterion.

The different colors in the performance figure indicate how the dimensionality of the parameter space (number of free parameters) strongly drives the inversion time. As the complexity of the model increases, the “volume” of the prior space grows, requiring more iterations to isolate the high-likelihood regions.

In addition, we also provide a comparison between PyMultiNest and nestle. While nestle is a convenient pure-Python implementation, PyMultiNest (leveraging the Fortran MultiNest library) generally demonstrates superior scaling and efficiency, especially as the number of free parameters increases beyond three.

To optimize the integration time of a **ForMoSA** inversion, we advise users to:

- Start low and run initial tests with a reduced number of live points (<100) to verify the model setup.
- Use a spectral resolution that matches the physical information content of the data; over-sampling the spectrum increases the inversion time without improving accuracy.
- Select the right algorithm for the inversion. For high-dimensional fits (>5 parameters), PyMultiNest is the recommended default.

Conclusions

ForMoSA is an active project that has been used in multiple published projects of research (see Appendix C). It will continue to evolve over time, with new functionalities being added to meet the requirements of future instruments (ELTs, HWO, etc.), updates to the included model grids, and the coupling of atmospheric models with additional physics to better fit the data (e.g., disks, extinction laws, multiple dimensions, time variability). Therefore, the information contained in this paper may change in the future. For this reason, the official documentation at ReadTheDocs, which will be regularly updated, should be considered the reference description of the code. As **ForMoSA** aims to provide the community with a tool that meets both current and future needs, the development team strongly encourages all users to provide feedback and suggestions via the GitHub issues tool in order to improve the code.

Authors contributions

Simon Petrus, Paulina Palma-Bifani, Matthieu Ravet, Allan Denis, and Bhavesh Rajpoot contributed equally to this work. The author order reflects the chronological order in which they joined the project. These authors developed the core functionalities, wrote the documentation and tutorials, and utilized **ForMoSA** to address diverse scientific questions regarding a variety of exoplanets and brown dwarfs.

Mickaël Bonnefoy and Gaël Chauvin have served as the main coordinators and supervisors of the team since the project’s start, guiding the first four authors who were doctoral students during the development of **ForMoSA**.

Arthur Vigan joined the project alongside Allan Denis in the context of HiRISE observations and the development of the High-Contrast High-Resolution (HCHR) module. Kevin Hoy joined

the project in the context of his HiRISE observing program and has served as a primary tester and debugger for the HCHR module.

Alice Radcliffe contributed by computing a new version of the Exo-REM model grid and testing its performance using **ForMoSA**, which also required the implementation of new features within the code.

AI usage disclosure

We did not rely on AI for the development of the code itself. All core architecture, logic, and implementation were designed and written by our team. AI was used solely as a supporting tool for documentation management and as an auxiliary resource during debugging.

Acknowledgements

The authors express their sincere thanks to the Code/Astro Workshop³, which provided the foundational training necessary to transform **ForMoSA** into a professional, open-source Python package.

We gratefully acknowledge the funding and support for the ForM-X workshops held in Nice (2023), Heidelberg (2024/2025), and Grenoble (2025). These collaborative sessions were instrumental in the development and refinement of the code. We also thank the various laboratories and institutions, especially IPAG, Lagrange, and MPIA, for their continued support.

Furthermore, this work has been supported by the French National Research Agency (ANR) through the MIRAGES project (PI: A. Vigan, ANR-20-CE31-0017).

We acknowledge support in France from the French National Research Agency (ANR) through project grant ANR-20-CE31-0012.

S. Petrus was supported by an appointment to the NASA Postdoctoral Program at the NASA-Goddard Space Flight Center, administered by Oak Ridge Associated Universities under contract with NASA.

Appendix A: State of the field on developments of atmospheric modeling tools

The field of exoplanetary atmospheric modeling has seen a significant increase in the availability of tools, largely divided between retrieval frameworks, which parameterize the atmospheric structures to fit the data, and forward modeling codes, which utilize self-consistent parametrized grids. While not intended as an exhaustive review of every available resource, Table 3 summarizes several of the most classical tools currently employed by the community. By situating **ForMoSA** within this broader ecosystem, we highlight its specific role as a modular, grid-based framework optimized for the next generation of substellar atmospheric analysis.

Table 3: Comparison of publicly available atmospheric modeling codes

Name & Reference	Description
CHIMERA	

³Code/Astro Workshop: <https://semaphore.github.io/codeastro/>

Name & Reference	Description
Line et al. (2013)	CHIMERA (Caltech Inverse ModEling and Retrieval Algorithms) is an open-source Python 3 package designed for the atmospheric retrieval and forward modeling of exoplanets. It interprets transmission and emission spectra from various instruments to constrain atmospheric properties. The code employs a chemically consistent framework or a free retrieval approach to estimate temperature profiles, gas abundances, and cloud properties. It utilizes high-performance acceleration via numba, supports multiple Bayesian samplers (e.g., PyMultiNest, Dynesty), and handles radiative transfer using correlated-K opacity treatments and two-stream approximations for scattering.
CROCODILE Hayoz et al. (2023)	CROCODILE (Cross-correlation Retrievals of Directly Imaged self-Luminous Exoplanets) is a Python-based statistical framework designed for the atmospheric characterization of close-in directly imaged exoplanets. It integrates different observing techniques (photometry, low-resolution spectroscopy, and medium-resolution cross-correlation spectroscopy) into a single robust Bayesian retrieval process. By handling the complexities of stellar speckle contamination, the code enables more accurate constraints on atmospheric thermal and chemical properties. It relies on the petitRADTRANS radiative transfer package and PyMultiNest for sampling.
Exo_k Leconte (2021)	Exo_k is a Python 3 library designed to handle radiative opacities for atmospheric applications. It provides tools to efficiently interpolate, convert, and adapt correlated-k and cross-section tables from a wide variety of sources and formats (e.g., Exomol, Nemesis, TauREx). Beyond opacity management, the library features an integrated radiative transfer framework and a full-fledged 1D atmospheric evolution model capable of computing the physical state of planetary atmospheres in radiative-convective equilibrium.
NEMESIS Irwin et al. (2008)	NEMESIS (Non-linear optimal Estimator for MultivariatE spectral analySIS) is a general-purpose radiative transfer and retrieval package designed for the characterization of planetary atmospheres, applicable to both Solar System bodies and exoplanets. It consists of two primary components: the retrieval tool itself and the underlying RADTRAN radiative transfer code. The framework is highly versatile, supporting a wide range of observing geometries and capable of operating in both correlated-k and line-by-line modes to analyze diverse spectral data.
petitRADTRANS	

Name & Reference	Description
Mollière et al. (2019)	petitRADTRANS is a versatile Python package that solves the radiative transfer equation for both transmission and emission geometries, featuring an updated opacity database that includes various isotopologues. Incorporating both low-resolution correlated-k and high-resolution Line-by-Line treatments, it enables the modeling of exoplanet atmospheres across a wide range of temperatures and pressures. Furthermore, it includes a robust integrated retrieval module that utilizes nested sampling to infer atmospheric parameters, such as chemical abundances and temperature profiles, directly from observed spectra.
SEDA Suarez et al. (2025)	SEDA (Spectral Energy Distribution Analyzer) is an open-source Python package designed for the forward modeling and empirical analysis of ultracool objects, including brown dwarfs, giant exoplanets, and low-mass stars. It facilitates comparisons between observed spectro-photometric data and grids of atmospheric models. The toolkit operates within a Bayesian framework to sample posteriors or utilizes chi-square minimization to determine best-fit models, while also providing data visualization functions.
species Stolker et al. (2020)	species is a toolkit designed for the spectral and photometric analysis of directly imaged exoplanets and brown dwarfs. It creates a unified framework that integrates observational data with various models and analysis tools, including flux calibration, spectral classification, analysis of evolutionary models, interpolation of model grids, parameter inference, and the creation of color-magnitude diagrams.

The codes mentioned in Table 3 are just a few examples; many others exist. Notably, APOLLO (Howe et al., 2017) is designed to model spectra and phase curves for transit spectroscopy. For high-resolution spectroscopy ($R_\lambda > 20,000$), ExoJAX (Kawahara et al., 2022) offers an auto-differentiable framework built on JAX. HOMER (Himes et al., 2022) focuses on 1D radiative-convective equilibrium, incorporating disequilibrium chemistry and photochemistry. For Bayesian inference, Pyrat Bay (Cubillos & Bleic, 2021) provides a versatile retrieval framework using MCMC and Nested Sampling. Finally, rfast (Robinson, 2023) is a high-speed, semi-analytical code optimized for the thermal and reflected light of rocky exoplanets.

Appendix B: Accuracy performances

Figure 3 illustrates the retrieval accuracy for the five parameters of the Exo-REM k26 grid (T_{eff} , $\log(g)$, $[M/H]$, C/O , and f_{sed}) as a function of spectral resolution, signal-to-noise ratio (S/N), and number of live points, following the same parameters explored in Figure 3.

The synthetic spectra simulate a K-band observation ($1.9\text{--}2.4\ \mu\text{m}$), which significantly limits the coverage of the spectral energy distribution (SED). At low resolution and low S/N , the impact is most pronounced for parameters sensitive to the SED shape (T_{eff} , $\log(g)$, and f_{sed}). Overall, above ~ 30 , the number of live points has minimal effect on the retrieval accuracy.

Accuracy comparison using PyMultiNest with varying spectral resolution (R_λ), signal-to-noise (S/N), and number of live points. Each dotted red line represents the expected value and black points the retrieved posteriors for each parameter explored during the nested sampling. The default setup is $R_\lambda = 368$, $S/N = 22$, and 215 live points. Each plot illustrates the effect of varying one parameter while keeping the others fixed.

Figure 3: Accuracy comparison using PyMultiNest with varying spectral resolution (R_λ), signal-to-noise (S/N), and number of live points. Each dotted red line represents the expected value and black points the retrieved posteriors for each parameter explored during the nested sampling. The default setup is $R_\lambda = 368$, $S/N = 22$, and 215 live points. Each plot illustrates the effect of varying one parameter while keeping the others fixed.

216 Appendix C: Research impact statement

217 Initiated in 2020, **ForMoSA** has since been employed for a variety of atmospheric analyses of
218 substellar objects, owing to its flexibility regarding model selection and data compatibility.
219 These works are based on different observations and also on different physical models.

220 To track the scientific output associated with this tool, we maintain a specific public library on
221 the NASA ADS platform⁴.

222 Here, we first showcase the standard usage of **ForMoSA** across different spectral resolutions,
223 ranging from medium ($R_\lambda < 10,000$) to low resolution and broadband photometry. We then
224 demonstrate the application of this method to large homogeneous target libraries observed and
225 extracted under similar conditions. Subsequently, we discuss results obtained by combining
226 heterogeneous datasets, and finally, we present the latest developments extending **ForMoSA** to
227 analyze high-contrast, high-resolution ($R_\lambda > 10,000$) observations.

228 Hereafter, we provide a brief overview of the 13 works published in peer-reviewed journals, as
229 well as several recent submissions.

230 Standard usage

231 Medium resolution spectroscopy

232 S. Petrus et al. (2021) re-detected the planetary-mass companion HIP 65426 b in VLT/SINFONI
233 data using the molecular mapping approach, which has recently emerged as a promising
234 technique to filter stellar residuals in datacubes of planetary systems obtained with Integral-
235 Field Units such as VLT/ERIS or Keck/Oriris. They used **ForMoSA** to demonstrate that
236 molecular mapping, coupled with atmospheric models, was able to characterize the atmosphere
237 of HIP 65426 b.

238 S. Petrus et al. (2023) characterized the companion VHS 1256 b, observed with VLT/X-Shooter
239 ($R_\lambda \sim 8000$, $\Delta\lambda = 0.5\text{--}2.5 \mu\text{m}$). They fitted the data using the ATM0 model and identified
240 the limitations of the approach, especially the non-reproducibility of the data at wavelengths
241 below $1.0 \mu\text{m}$ and the non-physical radius retrieved. They also showed that the well-known
242 variability of this object could be traced via the T_{eff} estimated with **ForMoSA**.

243 Low resolution spectroscopy

244 Palma-Bifani et al. (2024) analyzed the low-resolution spectrum of the companion AF Lep
245 b obtained with VLT/SPHERE ($R_\lambda \sim 30$, $\Delta\lambda = 0.95\text{--}1.65 \mu\text{m}$). They used the Exo-REM
246 model for the fits and found atmospheric properties consistent with a young, cold, early-T
247 super-Jovian planet.

248 Photometry only

⁴ForMoSA public library: [NASA ADS](https://ui.adsabs.org/author/ForMoSA/)

Carter et al. (2023) successfully extracted seven photometric points ($\Delta\lambda = 2.5\text{--}15.5\ \mu\text{m}$) of HIP 65426 b, the first planetary-mass companion imaged with JWST/NIRCam + MIRI. They fitted these points using the BT-Settl models and found atmospheric properties consistent with previous studies.

Large population studies

The relatively short computational time required to perform a fit with **ForMoSA** enables the homogeneous analysis of large spectral libraries. This approach allows for comparative studies aimed at identifying and mitigating potential systematic errors introduced by the adopted models. Such a strategy improves the robustness of the results and reduces the risk of misinterpretation.

S. Petrus et al. (2020) compared VLT/X-Shooter spectra of nine warm brown dwarfs with the BT-Settl atmospheric models. They identified systematic discrepancies between the observations and the models that could be partially alleviated by introducing interstellar extinction (A_V) as a free parameter. This result points to an incomplete treatment of clouds, dust, and haze in the models, leading to biases in the retrieved colors, and consequently, in the inferred T_{eff} of the targets.

Palma-Bifani et al. (2025) characterized the atmospheres of 21 M5–L5 companions and isolated brown dwarfs observed with VLT/SINFONI ($R_\lambda \sim 4000$; $\Delta\lambda = 1.95\text{--}2.45\ \mu\text{m}$). They measured a decrease in T_{eff} of more than 500 K across the M/L transition. They also confirmed that the retrieved atmospheric properties are strongly model-dependent, making their physical interpretation challenging. Finally, they found no significant differences between the atmospheric properties of companions and those of isolated objects.

S. Petrus et al. (2025) analyzed the X-SHYNE library, which comprises 43 young ($<500\ \text{Myr}$), low-mass ($<20\ \text{MJup}$), and cold ($T_{\text{eff}} \sim 600\text{--}2000\ \text{K}$) isolated brown dwarfs and wide-separation companions observed with VLT/X-Shooter. They compared the physical properties predicted by evolutionary models with those retrieved using **ForMoSA** in combination with atmospheric models. One of the main results of the study is the identification of a discrepancy in the inferred surface gravity ($\log g$). This inconsistency is likely linked to the diversity of cloud coverage in the atmospheres of X-SHYNE objects, potentially driven by differences in viewing geometry (e.g., equator-on versus pole-on orientations). Overall, the objects in the X-SHYNE sample exhibit near-solar metallicities and C/O ratios, as expected for objects formed in isolation as failed stars.

A framework for combination of datasets

The spectra provided by a given instrument are inherently limited by its spectral resolution and wavelength coverage. To maximize the diversity of spectral information used to constrain an object's atmosphere during the fitting process, it can be advantageous to combine data acquired with different instruments. However, the resulting inhomogeneity of the combined dataset must be carefully accounted for in the interpretation of the results to avoid introducing biases.

Palma-Bifani et al. (2023) combined multiple spectroscopic datasets and photometric measurements to cover a broad fraction of the spectral energy distribution of the planetary-mass companion AB Pic b. They included J, H, and K band spectra ($\Delta\lambda = 1.1\text{--}1.4$, $1.45\text{--}1.85$, and $1.95\text{--}2.45\ \mu\text{m}$, respectively) obtained with VLT/SINFONI ($R_\lambda \sim 1500\text{--}4000$), L'-band spectroscopy from Magellan-AO/CLIO2 ($R_\lambda \sim 300$, $\Delta\lambda = 3.4\text{--}4.1\ \mu\text{m}$), as well as photometric measurements from HST/WFC3 ($\Delta\lambda = 0.53\text{--}0.92\ \mu\text{m}$) and Spitzer/IRAC ($\Delta\lambda = 3.6\text{--}8\ \mu\text{m}$). They confirmed that the derived atmospheric properties are both model- and wavelength-dependent, and that the associated uncertainties are likely underestimated.

Simon Petrus et al. (2024) extended the characterization of VHS 1256 b previously performed by S. Petrus et al. (2023) using **ForMoSA** by analyzing its spectrum obtained

with JWST/NIRSpec+MIRI ($R_\lambda \sim 1500\text{--}3500$, $\Delta\lambda = 1.0\text{--}18.0 \mu\text{m}$). They exploited the wavelength dependence of the derived parameters to quantify biases induced by the models and propagated this systematic contribution directly into the uncertainties of the derived atmospheric properties. This approach enabled a more robust characterization of VHS 1256 b. In addition, they used **ForMoSA** to estimate the pseudo-continuum at the location of the absorption feature at $10 \mu\text{m}$, which is not reproduced by current atmospheric models. This allowed them to measure its equivalent width and quantify the cloud content in the atmosphere of the object.

Houll   et al. (2025) presented an analysis of the first VLT/MATISSE observations of β Pic b. In this study, they combined VLT/MATISSE data obtained in the L and M bands ($R_\lambda \sim 500$, $\Delta\lambda = 2.75\text{--}5.0 \mu\text{m}$) with previous VLT/GRAVITY observations in the K band ($R_\lambda \sim 500$, $\Delta\lambda = 2.0\text{--}2.45 \mu\text{m}$). Using **ForMoSA**, they fitted the two datasets both independently and jointly, and found that the derived atmospheric parameters vary depending on the fitting configuration adopted.

M. Ravet et al. (2025) developed the MOSAIC module within **ForMoSA** to mitigate biases arising from the combination of independent datasets obtained with different instruments. The method consists of computing an individual likelihood for each dataset and then combining them into a meta-likelihood, after introducing a scaling parameter, α , retrieved during the fit to account for potential inter-calibration offsets. They applied this strategy to an inhomogeneous dataset composed of spectra and photometric measurements of β Pic b obtained with ten different instruments. Their results demonstrate the importance of analyzing independent datasets with an adapted statistical framework in order to avoid misinterpretation of the derived atmospheric properties.

Matthieu Ravet et al. (2026) characterized the cold companion COCONUTS-2 b (T9) by combining photometric measurements from Spitzer (3.6 and $4.5 \mu\text{m}$) and WISE W1 ($3.32 \mu\text{m}$), W2 ($4.56 \mu\text{m}$) with spectra obtained from FLAMINGOS-2 ($R_\lambda \sim 900$, $\Delta\lambda = 1.01\text{--}2.50 \mu\text{m}$), JWST/NIRSpec ($R_\lambda \sim 3000$, $\Delta\lambda = 2.88\text{--}5.14 \mu\text{m}$), and JWST/MIRI-LRS ($R_\lambda \sim 100$, $\Delta\lambda = 5.45\text{--}11 \mu\text{m}$). Using the MOSAIC module of **ForMoSA**, they were able to robustly characterize the atmosphere of this object. They also demonstrated that classical noise-scaling schemes are poorly suited for heterogeneous, multi-modal datasets with varying spectral resolution and signal-to-noise ratio. In contrast, parametric covariance models provide a more reliable alternative for low- to moderate-sized observations.

330 The high-contrast high-resolution (HCHR) module

Denis et al. (2025) compared VLT/HiRISE data ($R_\lambda \sim 140,000$, $\Delta\lambda = 1.43\text{--}1.77 \mu\text{m}$; Vigan et al. (2024)) of AF Lep b with the Exo-REM atmospheric model. They were able to precisely measure the radial velocity of the companion to robustly constrain its orbit. They also confirmed the chemical abundances estimated from previous studies.

Denis et al. (2026) extended the methodology previously developed for AF Lep b to the colder planet 51 Eri b (~ 600 K), enabling the measurement of its radial velocity and revealing the complexity of its orbit.

338 Extended use cases of ForMoSA

Hoch et al. (2025) used **ForMoSA** to model the circumplanetary disk (CPD) detected around the exoplanet YSES 1 b. They combined the Exo-REM atmospheric model with a simple CPD model consisting of a single blackbody component. From this analysis, they derived a disk effective temperature $T_{\text{eff}} = 371 \pm 50$ K and a radius of 7.35 ± 2.25 RJup, consistent with results obtained using the parametric model *petitRADTRANS*.

Radcliffe et al., (accepted) implemented a two-column approach within **ForMoSA** to improve the fitting performance of heterogeneous atmospheres using 1D atmospheric models. This strategy consists of calling the same grid of precomputed synthetic spectra twice at each iteration and

combining them into a single synthetic spectrum using a weighting coefficient treated as a free parameter. This approach effectively simulates a heterogeneous atmosphere composed of two types of cloud patches distributed across the surface. They tested this method using a new version of the Exo-REM grid, which now includes the sedimentation factor f_{sed} as a free parameter. The dataset analyzed was the JWST spectrum of VHS 1256 b. Their results show a significant improvement in fit quality compared to previous analyses.

References

- Carter, A. L., Hinkley, S., Kammerer, J., Skemer, A., Biller, B. A., Leisenring, J. M., Millar-Blanchaer, M. A., Petrus, S., Stone, J. M., Ward-Duong, K., Wang, J. J., Girard, J. H., Hines, D. C., Perrin, M. D., Pueyo, L., Balmer, W. O., Bonavita, M., Bonnefoy, M., Chauvin, G., ... Zhang, Z. (2023). The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems I: High-contrast Imaging of the Exoplanet HIP 65426 b from 2 to 16 μm . *951*(1), L20. <https://doi.org/10.3847/2041-8213/acd93e>
- Cubillos, P. E., & Blecic, J. (2021). The PYRAT BAY framework for exoplanet atmospheric modelling: a population study of Hubble/WFC3 transmission spectra. *505*(2), 2675–2702. <https://doi.org/10.1093/mnras/stab1405>
- Denis, A., Vigan, A., Chauvin, G., Lacquement, A., Beust, H., Ravet, M., Costes, J., Radcliffe, A., Martos, S., Balmer, W., Stolker, T., Palma-Bifani, P., Rajpoot, B., Simonnin, A., Petrus, S., Forveille, T., Janson, M., Reiners, A., Godoy, N., ... Zins, G. (2026). Eccentric and cool: A high-spectral-resolution view of 51 Eri b with VLT/HiRISE. *707*, L13. <https://doi.org/10.1051/0004-6361/202658862>
- Denis, A., Vigan, A., Costes, J., Chauvin, G., Radcliffe, A., Ravet, M., Balmer, W., Palma-Bifani, P., Petrus, S., Parmentier, V., Martos, S., Simonnin, A., Bonnefoy, M., Cadet, R., Forveille, T., Charnay, B., Kiefer, F., Lagrange, A.-M., Chiavassa, A., ... Zins, G. (2025). Characterization of AF Lep b at high spectral resolution with VLT/HiRISE. *696*, A6. <https://doi.org/10.1051/0004-6361/202453108>
- Hayoz, J., Cugno, G., Quanz, S. P., Patapis, P., Alei, E., Bonse, M. J., Dannert, F. A., Garvin, E. O., Gebhard, T. D., Konrad, B. S., & Sartori, L. F. (2023). CROCODILE. Incorporating medium-resolution spectroscopy of close-in directly imaged exoplanets into atmospheric retrievals via cross-correlation. *678*, A178. <https://doi.org/10.1051/0004-6361/202245752>
- Himes, M. D., Harrington, J., Cobb, A. D., Güneş Baydin, A., Soboczenski, F., O’Beirne, M. D., Zorzan, S., Wright, D. C., Scheffer, Z., Domagal-Goldman, S. D., & Arney, G. N. (2022). Accurate Machine-learning Atmospheric Retrieval via a Neural-network Surrogate Model for Radiative Transfer. *3*(4), 91. <https://doi.org/10.3847/PSJ/abe3fd>
- Hoch, K. K. W., Rowland, M., Petrus, S., Nasedkin, E., Ingebreetsen, C., Kammerer, J., Perrin, M., D’Orazi, V., Balmer, W. O., Barman, T., Bonnefoy, M., Chauvin, G., Chen, C., De Rosa, R. J., Girard, J., Gonzales, E., Kenworthy, M., Konopacky, Q. M., Macintosh, B., ... Zhang, Y. (2025). Silicate clouds and a circumplanetary disk in the YSES-1 exoplanet system. *643*(8073), 938–942. <https://doi.org/10.1038/s41586-025-09174-w>
- Houllé, M., Millour, F., Berio, P., Scigliuto, J., Lacour, S., Lopez, B., Allouche, F., Augereau, J.-C., Blain, D., Bonnefoy, M., Carillet, M., Chauvin, G., Leftley, J., Matter, A., Milli, J., Mollière, P., Nasedkin, E., Nowak, M., Palma-Bifani, P., ... Weigelt, G. (2025). The mid-infrared spectrum of β Pictoris b: First VLTI/MATISSE interferometric observations of an exoplanet. *704*, A182. <https://doi.org/10.1051/0004-6361/202453323>
- Howe, A. R., Burrows, A., & Deming, D. (2017). An Information-theoretic Approach to Optimize JWST Observations and Retrievals of Transiting Exoplanet Atmospheres. *835*(1), 96. <https://doi.org/10.3847/1538-4357/835/1/96>
- Irwin, P. G. J., Teanby, N. A., de Kok, R., Fletcher, L. N., Howett, C. J. A., Tsang, C.

- 395 C. C., Wilson, C. F., Calcutt, S. B., Nixon, C. A., & Parrish, P. D. (2008). The
396 NEMESIS planetary atmosphere radiative transfer and retrieval tool. *109*, 1136–1150.
397 <https://doi.org/10.1016/j.jqsrt.2007.11.006>
- 398 Kawahara, H., Kawashima, Y., Masuda, K., Crossfield, I., van den Bekerom, D., Kitzmann, D.,
399 Morris, B., Pannier, E., Nugroho, S., & Ishikawa, T. (2022). *ExoJAX: Spectrum modeling*
400 *of exoplanets and brown dwarfs*. Astrophysics Source Code Library, record ascl:2206.003.
- 401 Landman, R., Stolker, T., Snellen, I. A. G., Costes, J., de Regt, S., Zhang, Y., Gandhi, S.,
402 Mollière, P., Kesseli, A., Vigan, A., & Sanchez-López, A. (2024). β Pictoris b through
403 the eyes of the upgraded CRIRES+. Atmospheric composition, spin rotation, and radial
404 velocity. *682*, A48. <https://doi.org/10.1051/0004-6361/202347846>
- 405 Leconte, J. (2021). Spectral binning of precomputed correlated-k coefficients. *645*, A20.
406 <https://doi.org/10.1051/0004-6361/202039040>
- 407 Line, M. R., Wolf, A. S., Zhang, X., Knutson, H., Kammer, J. A., Ellison, E., Deroo,
408 P., Crisp, D., & Yung, Y. L. (2013). A Systematic Retrieval Analysis of Secondary
409 Eclipse Spectra. I. A Comparison of Atmospheric Retrieval Techniques. *775*(2), 137.
410 <https://doi.org/10.1088/0004-637X/775/2/137>
- 411 Madhusudhan, N. (2018). Atmospheric Retrieval of Exoplanets. In H. J. Deeg & J. A. Belmonte
412 (Eds.), *Handbook of exoplanets* (p. 104). https://doi.org/10.1007/978-3-319-55333-7_104
- 413 Mollière, P., Wardenier, J. P., van Boekel, R., Henning, Th., Molaverdikhani, K., & Snellen,
414 I. A. G. (2019). petitRADTRANS. A Python radiative transfer package for exoplanet
415 characterization and retrieval. *627*, A67. <https://doi.org/10.1051/0004-6361/201935470>
- 416 Palma-Bifani, P., Bonnefoy, M., Chauvin, G., Rojo, P., Baudoz, P., Charnay, B., Denis, A.,
417 Hoch, K., Petrus, S., Ravet, M., Simonnin, A., & Vigan, A. (2025). The planetary-
418 mass-limit VLT/SINFONI library: Spectral extraction and atmospheric characterization via
419 forward modeling. *701*, A51. <https://doi.org/10.1051/0004-6361/202554894>
- 420 Palma-Bifani, P., Chauvin, G., Bonnefoy, M., Rojo, P. M., Petrus, S., Rodet, L., Langlois,
421 M., Allard, F., Charnay, B., Desgrange, C., Homeier, D., Lagrange, A.-M., Beuzit, J.-L.,
422 Baudoz, P., Boccaletti, A., Chomez, A., Delorme, P., Desidera, S., Feldt, M., ... Zhang, Y.
423 (2023). Peering into the young planetary system AB Pic. Atmosphere, orbit, obliquity, and
424 second planetary candidate. *670*, A90. <https://doi.org/10.1051/0004-6361/202244294>
- 425 Palma-Bifani, P., Chauvin, G., Borja, D., Bonnefoy, M., Petrus, S., Mesa, D., De Rosa, R.
426 J., Gratton, R., Baudoz, P., Boccaletti, A., Charnay, B., Desgrange, C., Tremblin, P., &
427 Vigan, A. (2024). Atmospheric properties of AF Lep b with forward modeling. *683*, A214.
428 <https://doi.org/10.1051/0004-6361/202347653>
- 429 Petrus, S., Bonnefoy, M., Chauvin, G., Babusiaux, C., Delorme, P., Lagrange, A.-M., Florent,
430 N., Bayo, A., Janson, M., Biller, B., Manjavacas, E., Marleau, G.-D., & Kopytova, T. (2020).
431 A new take on the low-mass brown dwarf companions on wide orbits in Upper-Scorpius.
432 *633*, A124. <https://doi.org/10.1051/0004-6361/201935732>
- 433 Petrus, S., Bonnefoy, M., Chauvin, G., Charnay, B., Marleau, G.-D., Gratton, R., Lagrange,
434 A.-M., Rameau, J., Mordasini, C., Nowak, M., Delorme, P., Boccaletti, A., Carlotti,
435 A., Houllé, M., Vigan, A., Allard, F., Desidera, S., D'Orazi, V., Hoeijmakers, H. J., ...
436 Lavie, B. (2021). Medium-resolution spectrum of the exoplanet HIP 65426 b. *648*, A59.
437 <https://doi.org/10.1051/0004-6361/202038914>
- 438 Petrus, S., Chauvin, G., Bonnefoy, M., Tremblin, P., Charnay, B., Delorme, P., Marleau,
439 G.-D., Bayo, A., Manjavacas, E., Lagrange, A.-M., Mollière, P., Palma-Bifani, P., Biller,
440 B., Jenkins, J.-S., Goyal, J.-M., & Hoch, K. (2023). X-SHYNE: X-shooter spectra of
441 young exoplanet analogs. I. A medium-resolution 0.65–2.5 μm one-shot spectrum of VHS
442 1256–1257 b. *670*, L9. <https://doi.org/10.1051/0004-6361/202244494>

- 443 Petrus, S., Chauvin, G., Bonnefoy, M., Tremblin, P., Morley, C., Charnay, B., Suarez, G.,
 444 Gagné, J., Palma-Bifani, P., Denis, A., Ravet, M., Bayo, A., Bézard, B., Biller, B., Delorme,
 445 P., Faherty, J., Goyal, J.-M., Hoch, K., Hoy, K., ... Zurlo, A. (2025). X-SHYNE: X-Shooter
 446 spectra of young exoplanet analogs: II. Presentation and analysis of the full library. *701*,
 447 A208. <https://doi.org/10.1051/0004-6361/202554986>
- 448 Petrus, Simon, Whiteford, N., Patapis, P., Biller, B. A., Skemer, A., Hinkley, S., Suárez, G.,
 449 Palma-Bifani, P., Morley, C. V., Tremblin, P., Charnay, B., Vos, J. M., Wang, J. J., Stone,
 450 J. M., Bonnefoy, M., Chauvin, G., Miles, B. E., Carter, A. L., Lueber, A., ... Zhou, Y. (2024).
 451 The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems.
 452 V. Do Self-consistent Atmospheric Models Represent JWST Spectra? A Showcase with
 453 VHS 1256–1257 b. *966*(1), L11. <https://doi.org/10.3847/2041-8213/ad3e7c>
- 454 Ravet, M., Bonnefoy, M., Chauvin, G., Lacour, S., Nowak, M., Charnay, B., Tremblin, P.,
 455 Homeier, D., Morley, C., Fortney, J., Denis, A., Petrus, S., Palma-Bifani, P., Landman, R.,
 456 Parker, L. T., Houllé, M., Chomez, A., Worthen, K., Kiefer, F., ... Shangquan, J. (2025).
 457 Multimodal atmospheric characterization of β Pictoris b: Adding high-resolution continuum
 458 spectra from GRAVITY. *704*, A325. <https://doi.org/10.1051/0004-6361/202553885>
- 459 Ravet, Matthieu, Bonnefoy, M., Chauvin, G., Zhang, Z., Faherty, J. K., Voyer, M., Phillips,
 460 M. W., Tremblin, P., Kiman, R., Copeland, J., Mang, J. J., Morley, C. V., Kühnle, H.,
 461 Charnay, B., de Regt, S., Mollière, P., Petrus, S., Denis, A., Radcliffe, A., ... Henning, T.
 462 K. (2026). Panchromatic View of the Frigid Jovian Exoplanet COCONUTS-2 b. *arXiv*
 463 *e-Prints*, arXiv:2604.07176. <https://doi.org/10.48550/arXiv.2604.07176>
- 464 Robinson, T. D. (2023). *rfast: Planetary spectral forward and inverse modeling tool*.
 465 Astrophysics Source Code Library, record ascl:2306.028.
- 466 Ruffio, J.-B., Macintosh, B., Konopacky, Q. M., Barman, T., De Rosa, R. J., Wang, J. J.,
 467 Wilcomb, K. K., Czekala, I., & Marois, C. (2019). Radial Velocity Measurements of HR 8799
 468 b and c with Medium Resolution Spectroscopy. *158*(5), 200. [https://doi.org/10.3847/1538-](https://doi.org/10.3847/1538-3881/ab4594)
 469 [3881/ab4594](https://doi.org/10.3847/1538-3881/ab4594)
- 470 Stolker, T., Quanz, S. P., Todorov, K. O., Kühn, J., Mollière, P., Meyer, M. R., Currie, T.,
 471 Daemgen, S., & Lavie, B. (2020). MIRACLES: atmospheric characterization of directly
 472 imaged planets and substellar companions at 4–5 μ m. I. Photometric analysis of β Pic b,
 473 HIP 65426 b, PZ Tel B, and HD 206893 B. *635*, A182. [https://doi.org/10.1051/0004-](https://doi.org/10.1051/0004-6361/201937159)
 474 [6361/201937159](https://doi.org/10.1051/0004-6361/201937159)
- 475 Suarez, G., Faherty, J., Cruz, K., Metchev, S., Kiman, R., Alejandro Merchan, S., Rothermich,
 476 A., Bardalez Gagliuffi, D., Whiteford, N., & Parrish, S. (2025). Spectral Energy Distribution
 477 Analyzer (SEDA) for Forward Modeling of Ultracool Objects. *American Astronomical*
 478 *Society Meeting Abstracts #245*, 245, 319.01.
- 479 Vigan, A., El Morsy, M., Lopez, M., Otten, G. P. P. L., Garcia, J., Costes, J., Muslimov, E.,
 480 Viret, A., Charles, Y., Zins, G., Murray, G., Costille, A., Paufigue, J., Seemann, U., Houllé,
 481 M., Anwand-Heerwart, H., Phillips, M., Abinanti, A., Balard, P., ... Valenzuela Soto, J.
 482 (2024). First light of VLT/HIRISE: High-resolution spectroscopy of young giant exoplanets.
 483 *A&A*, *682*, A16. <https://doi.org/10.1051/0004-6361/202348019>